Movement Quest

Project Owner	Details
Name	Deborah Maignan
Student ID	002327056
Date	17 October 2026
ClickUp Board	https://app.clickup.com/90141776384/v/l/2kydwpg0-334
GitHub	https://github.com/ma1gnan/MovementQuest_Project1.git
Branch	starter

1. **Proposed Group Information**

I, Deborah Maignan, am the sole member of this group. I am an undergraduate student pursuing a bachelor's degree in Computer Science.

Section	Deliverable
UI	Five Flutter screen roles, accessible navigation, stateful set cards
Data Layer	SQLite schema, repository methods, CRUD, Migrations
Testing	CRUD, correction/undo, offline and cold-restart evidence
Documentation	README, screenshots, ClickUp evidence, record walkthrough
Ownership	Deborah Maignan

2. **Project Summary**

Movement Quest will be designed to help Georgia State University Students plan short workout routines, record what activities they actually completed, and have personal progress they can reflect on. Through this app I am trying to solve the problem of heart disease, which is the leading cause of death in the United States, by implementing regular exercise at my university. There won't be competitive leadership boards in this version; it is primarily about personal growth and getting the momentum going. The old fashion paper planners are made better with this app because there is the ability to remove accidental checked sets or compare intended versus completed workout activities. My app emphasizes quick entry, transparent feedback, and reversible corrections.

Features of this app include users being able to create a workout plan, then opening that plan, which will be called a Quest, to log their individual sets, afterwards the user has the ability to undo, or edit the entry, next the user can review this session or any past sessions and total the completed sets up. All of the app's data will be stored locally on the device. This will be a Flutter application prototype with steadfast plans and logs, clear validation, and a reliable offline workflow.

Equal-scope contract mapping

Five screen roles	1.Workout Catalog/Dashboard, 2.Workout Create/Edit, 3. Workout Detail/Activity, 4. Training History/Insights, 5.Settings
Primary SQLite entity	Workout Plans: store workout name, exercise, equipment, target sets, target repetition, and intensity
Related SQLite entity	Training Logs: store linked workout ID, completed sets, actual repetitions, date, and reflection
CRUD operations	Create, View, Edit, and Delete
Stateful workflow	Start workout, record sets, correct or undo progress, finish/ reset workout, and restore saved records
Stateless widget	Reusable workout cards, label, progress indicators, and summary display
Stateful widget	Workout forms, intensity slider, set-by-set progress tracker, and interactive workout control
Numeric control	Intensity slider ranging from 1-10 and numeric input for sets and repetitions

Filter	Filter workout plans by equipment
Aggregate	Display the total number of completed workout sessions
Validation	Reject empty workout names, zero or negative repetitions, and invalid set counts
Delete behavior	Confirm before deleting a workout plan and explain that its linked training logs will also be deleted
Local smart helper	Rank the next workout plan using saved preferences and previous training logs
Testing	Verify CRUD, duplicate-log prevention, invalid inputs, offline operation, and persistence on two devices

Local Data Model and CRUD

WorkoutPlan:

id INTEGER PRIMARY KEY; title TEXT NOT NULL; exercise TEXT NOT NULL; target_sets INTEGER(1-10); target_reps INTEGER (1-30); notes TEXT;

Each training log represents one recorded set

TrainingLog:

id INTEGER PRIMARY plan_id INTEGER NOT NULL REFERENCES WorkoutPlan(id); performed_at TEXT; set_number INTEGER; actual_reps INTEGER; completed INTEGER (0/1); reflection TEXT

Plan CRUD : create on form, read on Dashboard, update on edit, delete with confirmation. Log CRUD: create when saving a set, read in active quest and history, update actual repetition/ completion/ reflection to create an entry, delete any mistake set with undo before finalizing. Undo must restore the previous value or deleted row; avoid duplicate logs by using stable IDs and transactions.

Required AI Learning component

Movement Quest will include a local smart helper called “Next Quest Recommendation Helper”. Users will benefit from an suggestion for next quest target repetitions based on the recently completed sets and the set user preferences priorly saved.

Inputs will include: preferred workout intensity (1-10), Available equipment: (none, dumbbells, resistance bands), Planned workout intensity (1-10), Previously completed workout sessions (count), Days since last completed workout (days)

The outputs will include: target reached, consider one more rep, consider one less rep; along with a short explanation. Next Quest Recommendation helper will not have the access to edit only to provide suggestions.

Proposed rule: if at least three recently completed sets exist and all meet or exceed target, suggest +1 rep up to 30; if at least three exist and two or more are at least 2 reps below target, suggest −1 rep down to 1; otherwise keep target. This helper is more for trial and error suggestions to offer the student motivation to keep the momentum of daily movement. This helper's baseline is to always keep the target's baseline. A counterexample is to consider: fixed weekly +1 reps; rejected because it ignores actual entries and can suggest increases after incomplete sessions.

Example:

“Recommended Quest: Full Body Basics - This workout matches your equipment and preferred intensity and has not been completed recently.”

3. **GitHub Repository Link**

This project active link is **GitHub:** https://github.com/ma1gnan/MovementQuest_Project1.git ; the working branch is titled starter

4. **Signed Statement Attachment (PDF or Clear Image)**

MOBILE APPLICATION DEVELOPMENT / PROJECT ONE

Team Project Acknowledgment

Attach a completed, signed copy to the proposal and include it in the final package.

Course / section
CSC 4360

Team name
Team Deb

Project title
Movement Quest

Acknowledgment of project requirements

By signing below, each listed member acknowledges the published Project One requirements and agrees to participate meaningfully throughout the group project.

1. The proposal is due October 17, 2026, and the final project is due October 27, 2026, both at 11:59 PM in Eastern Time (America/New_York). Instructor-team approval is required before implementation.
2. The app uses SQLite CRUD for both related entities, SharedPreferences settings, and one required explainable local smart helper with pathway-specific evaluation. Rules qualify when labeled honestly. No cloud storage, backend, authentication, or remote inference.
3. Each member contributes authentic work, maintains steady meaningful commits and ClickUp task-to-build evidence, and can explain their contribution, code, and decisions. AI assistance, if used, must be disclosed and verified under the assignment guidance.
4. The team provides a 20-25 minute presentation with integrated curated Q&A and evidence of app behavior, code, testing, and individual ownership.
5. Assessment is individual; members may receive different grades. Each member uploads the required final package individually to the designated iCollege destination.

This acknowledgment summarizes the existing assignment; it does not add new grading policies or replace the full project requirements.

Member signatures

All participating members must sign and date. Use the continuation area if needed.

Member 1 full name	Student ID	Signature	Date
Deborah Maignan	002327056		17 October 2026

Member 2 full name	Student ID	Signature	Date

Member 3 full name	Student ID	Signature	Date

5. Proposed Wireframe (UI Visual Plan)

